

Certified Integral Properties of the Dipper Block Cipher: Monomial-Trail Analysis and the Role of the Half-State Modular Addition

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Abstract

Dipper is a 64-bit lightweight block cipher whose round combines the GIFT S-box and bit permutation with two 16-bit modular additions on half of the state. Its integral resistance was supported only by experiments with a few random keys, which we replace by key-independent results. We model monomial trails through reduced-round Dipper as SAT instances with exact local transitions, including a rule for the modular additions derived from the Braeken–Semaev characterisation, and back the results by DRAT proofs and independently checked trails. The published five-round properties hold for all round keys, and six-round properties exist; the smallest certified six-round cube, of size 2^{60} , is minimal within the model. At seven rounds no bit-aligned cube is certified. Presence proofs, i.e. key monomials with an odd number of trails, show that this is exact for independent round keys and, more strongly, that no nonzero linear combination of output bits is balanced after seven rounds over any bit-aligned cube. They also show that 46 bits whose sums stayed zero in thousands of random-key trials are not balanced, and they determine the algebraic degree exactly in 1843 of 1856 (cipher, round, bit) cases. Replacing the two modular additions by XOR, or removing them, extends the longest certified property from six to nine and ten rounds. At the six-round boundary every certified bit comes from the two retained words, which pass through the additions unchanged. On all instances examined, the exact model certifies the same bits as the bit-based division property.

Keywords: integral cryptanalysis; monomial prediction; division property; modular addition; SAT; lightweight block cipher

1 Introduction

Integral (square, saturation) cryptanalysis [29] looks for a set of chosen plaintexts, typically an affine subspace called a *cube*, over which some output bit sums to zero independently of the key. The division property [38] and its bit-based versions [40, 45] turned the search for such properties into a propagation problem for MILP or SAT solvers. Monomial prediction [22], which Hu et al. proved equivalent to the three-subset division property without unknown subset [17], gives the exact algebraic view: a cube sum vanishes if and only if certain monomial coefficients are zero.

Modular addition is well understood in isolation. Its differential and linear behaviour is described exactly by Lipmaa and Moriai [32] and by Wallén [41], and its algebraic normal form

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by Braeken and Semaev [9]. Integral analysis of ciphers that mix modular addition with S-boxes is less developed. Division-property models for ARX designs [36, 35] target pure ARX ciphers such as SPECK [2], in which the addition is the only nonlinear operation. In a hybrid round the addition interacts with a small S-box and a bit permutation. It is then not obvious how much the addition contributes to integral resistance, or which part of the state keeps an integral property longest.

Dipper [26] is a 64-bit hybrid SPN–ARX cipher with 96- and 128-bit keys and 28 rounds, proposed in our previous work. Each round applies a full-state key addition, sixteen GIFT S-boxes [1], four word rotations, two 16-bit modular additions over half of the state and the GIFT-64 bit permutation. The design paper evaluates differential, linear and impossible-differential resistance with MILP and CP-SAT models. Its integral evaluation, however, is *experimental*: one 16-bit word is saturated, the sums are tested for a few random keys, and distinguishers are reported for up to five rounds. Such an evaluation has two weaknesses. A sum that is zero for a few random keys need not be zero for all keys, and a search limited to word-saturated cubes says nothing about other cubes. Section 6 shows that the first weakness is not hypothetical for Dipper: with 80 trials, two bits look balanced that are not.

Research questions. We ask three questions. (Q1) Which integral properties of reduced-round Dipper hold for *all* round keys, and how far do they extend over *all* bit-aligned cubes? (Q2) What exactly do the two half-state additions contribute, and why do some parts of the state stay balanced longer than others? (Q3) Does an exact treatment of the modular addition, made possible by the Hu–Yap characterisation [25], certify more than the conventional division property?

Contributions.

1. **Key-independent certificates and a complete search within the model (Q1).** We certify the published five-round properties for all round keys, find six-round properties (nine balanced bits with a 2^{63} cube; six bits with a cube of size 2^{60} , the smallest cube that the model certifies), and show that no bit-aligned cube certifies a single output bit at seven rounds in the model. The monotonicity lemma (Lemma 3) reduces this statement to the 64 cubes of dimension 63. Presence proofs (Section 4.3) show that this answer is exact for independent round keys: for each of the 64 cubes of dimension 63, every output bit is unbalanced after seven rounds, so by Lemma 6 no bit-aligned cube gives a seven-round property on a single output bit, for independent round keys. Section 6.4 extends this to every nonzero linear combination of output bits. The boundary results come with checkable artifacts (Section 5): DRAT proofs verified by `drat-trim` for the certificates, the degree bounds and every solver answer behind the presence proofs, including proofs that each trail enumeration is complete, and monomial trails validated by a checker that does not use the SAT encoding. The artifacts check the solver, not the CNF generator, whose local rules we verified exhaustively for small word sizes.
2. **The role of the modular addition (Q2).** In a controlled ablation, replacing the two additions by XOR or removing them extends the longest certified property from six to nine and ten rounds. At the six-round boundary every certified bit comes from the retained operands, the two words that feed the additions but pass through them unchanged. We relate this to Lemma 2, which states that output bit i of an addition has degree exactly $i + 1$, and to the algebraic degree of every output bit and round. We determine it exactly in 1843 of 1856 cases, by matching presence proofs to the certified upper bounds.
3. **Model comparison and tightness (Q3).** The monomial-trail existence model with exact local transitions (MP-EL), a gate-level monomial model and the conventional bit-based division property certify the same bits on every instance we examined. We explain

part of this structurally. We then compare certificates with experiment on 112 (cube, round) instances and use presence proofs to decide every bit that experiment leaves open. The 46 bits that stayed zero in every trial without being certified are all unbalanced. On these instances the certificates are therefore exactly the balanced bits, and a model that counts trails could not certify more.

Behind these results are four technical tools: the exact local rule for Dipper’s ARX map (Lemma 1), the standard free-final-round observation (Lemma 4), presence proofs, which count trails under maximal key patterns (Section 4.3), and a last-round component lemma (Lemma 8). The lemma formalises, for Dipper, the last-round choice of key monomials of Hebborn et al. [19]; we use it to turn the low-degree components of the GIFT S-box from an obstacle for counting into exact matrix entries. Sections 9 and 10 contain secondary material: a partial-key filtering procedure and two corrections to the key-schedule description of [26]. Code, data, proofs and scripts are available at <https://github.com/AlexGuseinov/dipper-integral>.

Scope of the claims. A certificate proves that a cube sum is zero for all round keys. The absence of a certificate proves nothing about the cipher: it only means that the model contains a trail. Where we claim more, we use presence proofs: an odd number of trails for one key monomial proves that the cube sum is a nonzero polynomial in the round keys, so that the bit is not balanced. With presence proofs for all 64 cubes of dimension 63, our seven-round statement holds for Dipper with independent round keys, not only within the model. We do not claim it for round keys produced by the key schedule. For cipher outputs we call a degree exact only when a presence proof matches the certified upper bound, and otherwise report the interval between the two.

Organisation. Section 2 reviews related work and Section 3 introduces the notation and background. Section 4 gives the local rules and the structural lemmas, and Section 5 validates them. Section 6 reports the certified properties of Dipper, and Section 7 studies the role of the addition. Section 8 compares models and experiment. Sections 9 and 10 cover key filtering and the key-schedule corrections. Section 11 discusses implications and limitations.

2 Related work

Integral and higher-order differential cryptanalysis. Integral attacks go back to the Square attack [11]. They were formalised as integral cryptanalysis by Knudsen and Wagner [29] and as the saturation attack by Lucks [33], and extended to bit-oriented ciphers by Z’aba et al. [46]. The underlying algebra is that of higher-order derivatives [31, 28]: a cube sum is a derivative whose order is the cube dimension, and it vanishes whenever the algebraic degree is smaller than that order. Canteaut and Videau [10] and Boura, Canteaut and De Cannière [8] bounded the degree of iterated constructions. The cube attack of Dinur and Shamir [13] applies the same algebra to key recovery.

Division property and monomial prediction. Todo introduced the division property [38] and used it to break full MISTY1 [37]. The bit-based division property [40] and its MILP modelling [45] made automated search practical, and Eskandari et al. [14] provide a SAT-based tool and a table of results that we use as a benchmark. In stream ciphers the division property underlies modern cube attacks [39, 43]. The three-subset division property [23] and its variant without unknown subset [17] remove the imprecision of the two-subset version; Hu et al. [22] showed that the latter is equivalent to monomial prediction. Complex linear layers need dedicated models [24, 48], and the precision of the division property can be increased by propagating through larger S-box layers [12]. Hebborn, Leander and Udovenko give a mathematical treatment of the conventional and the perfect division property [20].

Modular addition. Sun, Wang and Wang [36] gave the first MILP model of the bit-based division property for ARX ciphers, based on a gate-level decomposition of the carry chain, and Sun et al. [35] moved it to SAT. Hu and Yap [25] observed that the characterisation of Braeken and Semaev [9] yields a perfect local monomial-prediction model for modular addition. CLAASP-MP [4] integrates such models into a general MILP framework. We use this characterisation for the specific map $(x, y) \mapsto (x \boxplus y, y)$ of Dipper and derive from it a degree statement.

Security guarantees and degree lower bounds. Finding an integral distinguisher answers an upper-bound question. Proving that *no* integral distinguisher exists requires exact information, such as lower bounds on the degree. Hebborn et al. developed degree lower bounds [18] and strong security guarantees against integral distinguishers [19] for ciphers in which a key is added to the full state; Dipper adds a full-state key before every S-box layer and meets this assumption, but its round also contains a nonlinear ARX layer (Section 11). Zeng and Tian [47] extended the guarantees to ciphers without such a whitening key, such as SIMON and Simeck. Peng et al. [34] identified *interfering monomials*, which make trail counts even for badly chosen key monomials, and avoided them by a careful choice of the key monomials. Wang, Hadipour and Gerhalter [42] search for key-independent and weak-key integral combinations beyond single output bits. Beierle et al. [3] treated key whitening by modular addition, and Gerhalter and Eichlseder [15] complex linear layers. In all these works the proof of *presence* of a monomial, by an odd number of trails, is the basic step; Section 4.3 applies it to Dipper.

Integral key recovery. Hadipour and Eichlseder [16], for example, study key recovery built on monomial prediction.

Positioning. We do not propose a new general framework. Our contribution is the analysis of one hybrid cipher, which consists of exact rules for its particular ARX shape, key-independent certificates with checkable artifacts, a search over all bit-aligned cubes within the model, a model comparison, an ablation, and a degree-based explanation.

3 Preliminaries

This section fixes notation and recalls the algebraic facts on which the analysis rests. The material is standard, but we state it in the form in which we use it, because the difference between what a model *proves* and what it merely *fails to exclude* runs through the whole paper.

3.1 Notation

Vectors in $\mathbb{F}_2^n$ are written $u = (u_0, \dots, u_{n-1})$ with u_0 the least significant bit. We identify u with the integer $\text{val}(u) = \sum_i u_i 2^i$ whenever we do arithmetic with it. The Hamming weight is $\text{wt}(u) = \#\{i : u_i = 1\}$, and e_i is the i -th unit vector. We write $u \preceq v$ if $u_i \leq v_i$ for all i , so that u is a sub-mask of v ; $u \vee v$ and $u \wedge v$ are the bitwise OR and AND. For a set $I \subseteq \{0, \dots, n-1\}$, $\mathbf{1}_I$ is its indicator vector. The symbol $\oplus$ denotes XOR, and $\boxplus$ addition modulo 2^n , where n is clear from the context. $x \lll s$ is a left rotation of an n -bit word.

3.2 Boolean functions, ANF and degree

Every Boolean function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ has a unique algebraic normal form (ANF)

$$f(x) = \bigoplus_{u \in \mathbb{F}_2^n} a_u x^u, \quad x^u = \prod_{i: u_i=1} x_i, \quad a_u \in \mathbb{F}_2.$$

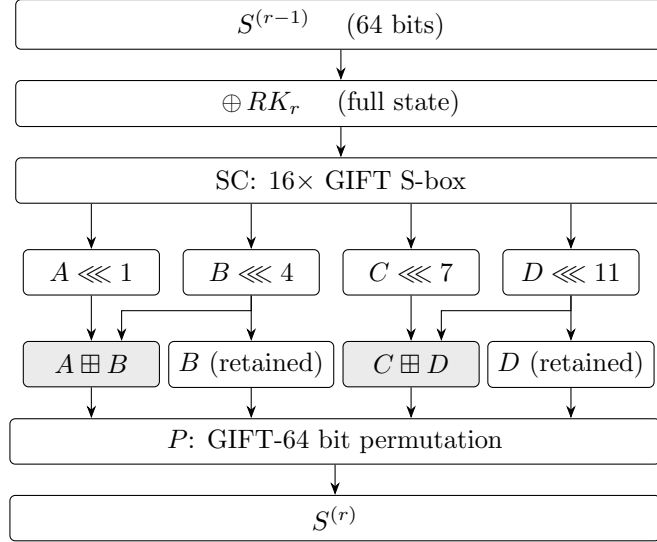


Figure 1: One Dipper round. The shaded boxes are the *added* words; B and D are the *retained* words, used as addends and passed on unchanged. The ARX layer M consists of the rotations and the two additions. There is no key addition after the last round.

The coefficients follow from the truth table by the Möbius transform $a_u = \bigoplus_{v \preceq u} f(v)$, which costs $n2^n$ operations. The *algebraic degree* $\deg f$ is the largest $\text{wt}(u)$ with $a_u = 1$. For a vectorial function $F = (f_0, \dots, f_{m-1})$ and $v \in \mathbb{F}_2^m$ we write $F^v = \prod_{j:v_j=1} f_j$, which is again a Boolean function. Two facts are used repeatedly: $x_i^2 = x_i$, so products of monomials are monomials of the union of their supports; and a nonconstant component of a permutation of $\mathbb{F}_2^n$ has degree at most $n - 1$, since $\bigoplus_x f(x)$ equals the coefficient a_1 and is zero for a balanced function.

3.3 The Dipper cipher

Figure 1 shows one round. The state is $S = A\|B\|C\|D$ with 16-bit words, $A = S[63:48]$, $D = S[15:0]$ and bit 0 the least significant. For $r = 1, \dots, 28$,

$$S^{(r)} = T(S^{(r-1)} \oplus RK_r), \quad T = P \circ M \circ SC,$$

with $S^{(0)}$ the plaintext and $S^{(28)}$ the ciphertext. SC applies the GIFT S-box to the sixteen nibbles. M first rotates A, B, C, D left by 1, 4, 7, 11 and then sets

$$A \leftarrow A \boxplus B, \quad C \leftarrow C \boxplus D,$$

so B and D are used as addends and pass through the additions unchanged (after their rotation). We call A, C the *added* words and B, D the *retained* words of the ARX layer. P is the GIFT-64 bit permutation $P(i) = 4\lfloor i/16 \rfloor + 16((3\lfloor (i \bmod 16)/4 \rfloor + (i \bmod 4)) \bmod 4) + (i \bmod 4)$, which moves bit i to position $P(i)$. Three structural features matter for integral analysis. The key is added to the *full* state at the start of every round. T is a public permutation. There is no key addition after the last round.

For the ablation we define two analysis variants, which are not ciphers anyone proposes. $\text{Dipper}^\oplus$ replaces both additions by XOR, and $\text{Dipper}^\emptyset$ removes them while keeping the rotations. Throughout, the round keys $RK_1, \dots, RK_r$ are treated as *independent* variables; a statement that holds for all such sequences holds in particular for the sequences produced by either key schedule.

3.4 Cubes and integral properties

Definition 1 (Cube, balanced bit). Let $I \subseteq \{0, \dots, 63\}$ and $c \in \mathbb{F}_2^{64}$ with $c_i = 0$ for $i \in I$. The *cube* $C_I(c)$ is the set of the $2^{|I|}$ plaintexts that take all values on the positions in I (the active bits) and agree with c elsewhere (the constant bits). For an r -round state bit $S_j^{(r)}$, the *cube sum* is $\bigoplus_{x \in C_I(c)} S_j^{(r)}(x)$. The bit is *balanced over I after r rounds* if the cube sum is zero for every constant c and every sequence of round keys.

All cubes in this paper are *bit-aligned*: they are spanned by unit vectors. Affine subspaces in other bases, non-affine input sets and output functions other than a single bit are outside the scope of our search.

Viewing $S_j^{(r)}$ as a polynomial in the plaintext bits x and the key bits k , and substituting the constants c_i for the plaintext bits outside I after summation,

$$\bigoplus_{x_I \in \mathbb{F}_2^{|I|}} S_j^{(r)}(x, k) = \bigoplus_{u \succeq \mathbf{1}_I, v} a_{u,v} x^{u \oplus \mathbf{1}_I} k^v, \quad (1)$$

because summing a monomial x^u over the cube leaves $x^{u \oplus \mathbf{1}_I}$ if $u \succeq \mathbf{1}_I$ and zero otherwise. Hence the bit is balanced over I *if and only if* every coefficient $a_{u,v}$ with $u \succeq \mathbf{1}_I$ vanishes. A cube sum is a derivative of order $|I|$ [31]. In particular, if $\deg_x S_j^{(r)} < |I|$ the bit is balanced over every cube of dimension $|I|$, which links integral properties to degree bounds (Section 7.2). The 64-dimensional cube, i.e. the full codebook, is balanced for every bit of every permutation. It is trivial and excluded.

3.5 Monomial trails

Let $F = F_r \circ \dots \circ F_1$ be a composition of vectorial functions. For a single function G and masks u, v , write $G[u \rightarrow v] = 1$ if x^u occurs in the ANF of G^v and 0 otherwise. A *monomial trail* from u_0 to u_r is a sequence $u_0 \rightarrow u_1 \rightarrow \dots \rightarrow u_r$ with $F_t[u_{t-1} \rightarrow u_t] = 1$ for all t .

Proposition 1 (Hu et al. [22]). *The coefficient of x^{u_0} in F^{u_r} equals the number of monomial trails from u_0 to u_r modulo 2.*

The proposition gives two sound tests of opposite strength. If *no* trail exists, the coefficient is zero; this is the test we use, and it only needs a satisfiability check. If trails exist, the coefficient is their parity. Deciding it requires counting, and trails can cancel in pairs. The following toy example shows the gap between the two tests.

Example 1 (Cancellation). Let $F_1(x_1, x_2) = (x_1 x_2, x_1 x_2)$ and $F_2(y_1, y_2) = y_1 \oplus y_2$. Then $F = F_2 \circ F_1 = 0$, so the sum over the cube $\{x_1, x_2\}$ is zero. From $x^{11} = x_1 x_2$ to the output there are two trails, $11 \rightarrow 10 \rightarrow 1$ and $11 \rightarrow 01 \rightarrow 1$, because $x_1 x_2$ occurs in both y_1 and y_2 and both occur in F_2 . An existence test finds a trail and cannot certify the property. The parity test counts two trails and certifies it.

For keyed functions we treat each key bit as a variable. The key-addition layer $s' = s \oplus k$ satisfies $s'^w = \bigoplus_{u \preceq w} s^u k^{w \oplus u}$. A trail step $u \rightarrow w$ is therefore possible exactly when $u \preceq w$, and different steps attach different key monomials $k^{w \oplus u}$. Trails with different key monomials never cancel each other; only trails with the same key monomial can.

3.6 The bit-based division property

The conventional (two-subset) bit-based division property [40] describes a multiset $\mathbb{X} \subseteq \mathbb{F}_2^n$ by a set $\mathbb{K}$ of vectors: $\mathbb{X}$ has property $\mathcal{D}_{\mathbb{K}}^n$ if $\bigoplus_{x \in \mathbb{X}} x^u = 0$ for every u such that $u \not\preceq k$ for all $k \in \mathbb{K}$. A cube with active set I has $\mathbb{K} = \{\mathbf{1}_I\}$. Propagation rules map $\mathbb{K}$ through each operation, and bit

j is balanced if e_j is not reachable. Key addition leaves the property unchanged. For an S-box, $k \rightarrow v$ is a *division trail* if some $u \succeq k$ has $S[u \rightarrow v] = 1$. The rules for COPY, AND and XOR gates are those of [40, 45]. BDP is sound in the same direction as the existence test above. Like that test, it cannot see cancellations, and it can be less precise still. Precision is improved by the three-subset division property [23] and made exact by its variant without unknown subset [17], which is equivalent to monomial prediction [22].

3.7 SAT encodings and checkable answers

Both tests become satisfiability problems once every mask bit is a Boolean variable and every local relation is written as clauses. Each clause excludes one forbidden assignment of the variables of a small component. The solver then answers one of two ways. If the formula is *unsatisfiable* (UNSAT), no trail exists and the bit is certified; a DRAT proof [44] records the solver’s reasoning, and an independent checker such as `drat-trim` can replay it. If the formula is *satisfiable* (SAT), the solver returns an assignment, from which we read off a candidate trail and verify it layer by layer against the local rules, without using the clauses. We use CaDiCaL [6, 5] through PySAT [27] for the search and an external CaDiCaL binary for proof generation.

4 Models

4.1 Local propagation rules

Key addition. A step $u \rightarrow w$ is allowed iff $u \preceq w$. Under BDP the key addition is transparent.

S-box. For the GIFT S-box we compute $S[u \rightarrow v]$ for all $u, v \in \mathbb{F}_2^4$ by the Möbius transform. Of the 256 pairs, 68 are nonzero. The BDP table has 170 transitions.

Modular addition with retained operand. Braeken and Semaev [9], and Hu and Yap [25] in the cryptanalytic setting, show that for $z = x \boxplus y$ on n bits the monomial $x^a y^b$ occurs in z^w if and only if $\text{val}(a) + \text{val}(b) = \text{val}(w)$ over the integers. Dipper’s ARX map also outputs the addend. The following lemma gives the exact rule for that shape.

Lemma 1. *Let $F(x, y) = (x \boxplus y, y)$ on n -bit words. Then $[x^a y^b] (x \boxplus y)^w y^c = 1$ if and only if $\text{val}(w) \geq \text{val}(a)$ and $b = b' \vee c$, where b' is the word with $\text{val}(b') = \text{val}(w) - \text{val}(a)$.*

Proof. By the Braeken–Semaev characterisation, $(x \boxplus y)^w = \sum_{\text{val}(a') + \text{val}(b') = \text{val}(w)} x^{a'} y^{b'}$. Multiplying by y^c and using $y_i^2 = y_i$ gives $\sum x^{a'} y^{b' \vee c}$. The coefficient of $x^a y^b$ is the parity of $\#\{b' : \text{val}(a) + \text{val}(b') = \text{val}(w), b' \vee c = b\}$. The equation determines b' uniquely, and b' exists iff $\text{val}(w) \geq \text{val}(a)$. $\square$

Example 2. For $n = 2$, the high output bit of $z = x \boxplus y$ is $z_1 = x_1 \oplus y_1 \oplus x_0 y_0$. Its monomials correspond to the pairs $(\text{val}(a), \text{val}(b)) \in \{(2, 0), (0, 2), (1, 1)\}$, which are exactly the solutions of $\text{val}(a) + \text{val}(b) = 2 = \text{val}(e_1)$. The monomial $x_0 y_0$ comes from the carry, and its degree $2 = i + 1$ anticipates Lemma 2 below. For the retained version, $z_1 y_0 = x_1 y_0 \oplus y_1 y_0 \oplus x_0 y_0$. With $w = e_1$ and $c = e_0$, Lemma 1 predicts the monomials $x^a y^{b' \vee e_0}$ for the same three pairs, i.e. $x_1 y_0$, $y_1 y_0$ and $x_0 y_0$, in agreement with the direct computation.

The same characterisation gives the exact degree of each output bit of the addition. Lemma 2 is an elementary consequence of [9]. We state it because it drives the mechanism of Section 7.2.

Lemma 2 (Degree of an addition bit). *Let $z = x \boxplus y$ on n -bit words. For $0 \leq i < n$, the output bit z_i has algebraic degree exactly $i + 1$ as a Boolean function of the $2n$ input bits.*

Proof. Take $w = e_i$, i.e. $\text{val}(w) = 2^i$. The monomial $x^a y^b$ occurs in z_i iff $\text{val}(a) + \text{val}(b) = 2^i$, and it has degree $\text{wt}(a) + \text{wt}(b)$. Every carry in the binary addition of a and b turns two ones into a single one in the next position, so $\text{wt}(a) + \text{wt}(b) = \text{wt}(a + b) + c(a, b)$, where $c(a, b)$ is the number of carries (equivalently, by Kummer's theorem [30], $c(a, b)$ is the 2-adic valuation of $\binom{a+b}{a}$). Here $\text{wt}(a + b) = 1$, and carries can occur only at positions $0, \dots, i - 1$, so the degree is at most $i + 1$. The pair $a = 1, b = 2^i - 1$ produces i carries and attains $i + 1$. $\square$

Corollary 1. *In one Dipper round, let d be an upper bound on the degree of the round input. Every bit of the retained words after the ARX layer is an S-box output bit and has degree at most $3d$. Bit i of an added word is a polynomial of degree $i + 1$ in S-box output bits, so its degree is at most $\min\{3d(i + 1), 63\}$. Only bit 0 of an added word is linear in the S-box outputs.*

Corollary 1 gives upper bounds; it does not state which degrees are attained.

Encoding. In the SAT model the equation $\text{val}(a) + \text{val}(b') = \text{val}(w)$ is encoded bitwise with auxiliary carries, $a_i + b'_i + q_i = w_i + 2q_{i+1}$, $q_0 = q_n = 0$. These are carries of the exponent integers, not the cipher's data carries. The pair (a, w) determines b' and q , so each trail has exactly one extension to the auxiliary variables.

The three models. MP-EL (monomial-trail existence model with exact local transitions) uses the S-box table and Lemma 1. Its local relations are exact, but its global test is trail *existence*, not trail parity, so it cannot see cancellations of the kind in Example 1. For comparison, MP-circuit models the addition as a ripple-carry circuit, $s_i = x_i \oplus y_i \oplus c_i$ and $c_{i+1} = x_i y_i \oplus (x_i \oplus y_i) c_i$, with explicit COPY gates, including the copy of y that forms the retained output. It uses the monomial gate rules (COPY: $u = \bigvee v_j$; AND: $u_1 = u_2 = v$; XOR: $v = u_1 + u_2$). BDP uses the same circuit with the division-property rules (COPY: $u = \sum v_j$; AND: $v = u_1 \vee u_2$) and the BDP S-box table.

4.2 Round model, soundness and structural lemmas

One round is modelled as the key addition, sixteen S-box relations, the word rotations (wiring), the two addition relations and the bit permutation (wiring). The input mask is fixed to $\mathbf{1}_I$ and the output mask after r rounds to e_j . Each output bit is an incremental query under assumptions.

Proposition 2 (Soundness). *If MP-EL (or BDP) is unsatisfiable for (I, r, j) , then bit j of $S^{(r)}$ is balanced over I : the cube sum is zero for every constant and every sequence of round keys, and in particular under both Dipper key schedules.*

Proof. Every local relation contains all transitions with nonzero coefficient. By Proposition 1, every nonzero coefficient $a_{u,v}$ with $u \succeq \mathbf{1}_I$ in (1) therefore has at least one trail starting from u . Such a trail first passes the key addition to some $w \succeq u \succeq \mathbf{1}_I$. Since $\mathbf{1}_I \preceq w$ is also a valid key-addition step, the model, whose input mask is exactly $\mathbf{1}_I$, contains a trail too. For BDP the statement is the standard soundness of division-property propagation [40, 45]. The round keys are independent variables and the constants merge with RK_1 , which gives the claimed generality. $\square$

Lemma 3 (Monotonicity). *In MP-EL, if $I \subseteq I'$ and (I, r, j) has no trail, then (I', r, j) has no trail. Consequently, if no bit-aligned cube of dimension 63 is certified for output bit j after r rounds, no bit-aligned cube of any dimension ≤ 63 is.*

Proof. The first operation is the key addition, so the masks reachable from $\mathbf{1}_I$ after it are $\{w \succeq \mathbf{1}_I\}$. This set contains $\{w \succeq \mathbf{1}_{I'}\}$, and all later constraints are identical. Every cube of dimension at most 63 lies inside some cube of dimension 63. $\square$

Lemma 4 (Free final round). *If bit j of $S^{(r)}$ is balanced over I and $|I| \geq 1$, then bit j of $T^{-1}(S^{(r+1)})$ is balanced over I .*

Proof. $T^{-1}(S^{(r+1)}) = S^{(r)} \oplus RK_{r+1}$, and the constant RK_{r+1} cancels over the even number $2^{|I|}$ of terms. $\square$

Lemma 4 uses a different output test: the ciphertext is first passed through the public map T^{-1} . It gives no information about RK_{r+1} . We report round counts for $S^{(r)}$ and state the $(r+1)$ -round extension separately.

Remark 1 (Degree bounds versus cubes). A bound $\deg_x S_j^{(r)} \leq 62$ certifies bit j over every 63-dimensional cube. The converse direction is weaker. Even a proof that $\deg_x S_j^{(r)} = 63$ would exclude a property for only those cubes I whose monomial x^{1_I} actually occurs. Excluding all 63-dimensional cubes would require the presence of the monomial x^{1_I} for each of the 64 choices of I separately.

4.3 Presence proofs

A certificate proves that monomials are absent. To prove that a bit is *not* balanced we need the opposite: one coefficient $a_{1_I, v}$ in (1) that equals 1. Here $v = (v_1, \dots, v_r)$ is a *key pattern*, with v_t the mask of the key monomial taken from RK_t . In a trail, the key-addition step $u_t \rightarrow w_t$ of round t attaches $k^{w_t \oplus u_t}$ (Section 3), so each trail has exactly one key pattern, namely $v_t = w_t \oplus u_t$.

Lemma 5 (Presence). *If the number of monomial trails from 1_I to e_j with key pattern v is odd, then bit j of $S^{(r)}$ is not balanced over I for independent round keys. More precisely, for the constant $c = 0$ the cube sum is a nonzero polynomial in the round keys, and for every constant c some sequence of round keys gives a nonzero cube sum.*

Proof. Treat the round-key bits as additional input variables. Each of them enters the cipher exactly once, through a key addition, so the trails of the extended function from $x^{1_I} k^v$ to bit j are exactly the trails above with key pattern v . By Proposition 1, $a_{1_I, v} = 1$. For $c = 0$, the right-hand side of (1) reduces to $\bigoplus_{v'} a_{1_I, v'} k^{v'}$, which contains k^v and is therefore not the zero polynomial. The constant enters only through $c \oplus RK_1$, so replacing RK_1 by $RK_1 \oplus c$ transfers a nonzero sum from $c = 0$ to any other constant. $\square$

Lemma 6 (Monotonicity of balancedness). *If bit j of $S^{(r)}$ is balanced over I , it is balanced over every $I' \supseteq I$. Equivalently, if it is not balanced over I' , it is not balanced over any $I \subseteq I'$.*

Proof. The cube $C_{I'}(c)$ is the disjoint union of the $2^{|I' \setminus I|}$ cubes $C_I(c')$, where c' runs over the constants that agree with c outside I' and are zero on I . Each of these cube sums is zero. $\square$

Lemma 6 is the cipher-level counterpart of Lemma 3. One presence proof for a 63-dimensional cube therefore excludes a property on that output bit for all 2^{63} bit-aligned cubes contained in it, i.e. all cubes that keep the same bit constant.

Search for a key pattern. The choice of v decides whether a count is feasible. A pattern with few key bits leaves many trails: in a first attempt we took the key patterns of 20 trails found by the solver on each of two seven-round instances, and none of them gave an odd count within $2 \cdot 10^4$ trails. Every key bit taken in round t forces the state mask to zero at that position and removes trails. We therefore use *maximal* patterns. Starting from the empty pattern, we visit the $64r$ key positions in random order and add a position whenever at least one trail survives. We then count the trails of the resulting pattern by enumerating the solutions of the MP-EL instance under assumptions, with the input mask, the output mask and the key pattern fixed by assumptions and with blocking clauses over the layer masks; all other variables are determined

by the masks and the pattern. A proof needs a small odd count, and patterns with large counts are useless, so the enumeration stops at a small cap (1500 trails, raised to 8000 and $4 \cdot 10^4$ for the few bits not resolved at the first stage), and many random restarts are made. A pattern that proved a bit for one cube is tried first for the same bit of the other cubes, as a whole and then with only its rounds 3–7 (or 4–7) kept and the earlier rounds searched afresh. For the 63-dimensional cubes we also fix $v_1 = 0$. Nothing is lost: the only other choice, $v_1 = e_p$, makes the state monomial after the first key addition the product of all 64 bits, and its trail count is even, because every output bit of the remaining rounds is a component of a permutation and has degree at most 63. For each proof the trail returned by the enumeration is validated by the independent checker of Section 5. The count is correct if every enumerated trail is valid, which we check for every trail, and if no trail is missed. The latter has two parts. That no solution of the CNF is missed is certified by a DRAT proof that the instance becomes unsatisfiable once the trails found are excluded (Section 5). That every trail is a solution of the CNF rests on the exactness of the local rules, which we verified exhaustively for small word sizes.

Two further lemmas extend presence proofs from single output bits to linear combinations of output bits. The first is the rank argument of Hebborn et al. [19], restricted to the single input monomial x^{1_I} .

Lemma 7 (Linear combinations). *Let $v^{(0)}, \dots, v^{(63)}$ be key patterns of r rounds, and let M be the 64×64 matrix over $\mathbb{F}_2$ whose entry $M_{l,j}$ is the number of monomial trails from $\mathbf{1}_I$ to e_j with key pattern $v^{(l)}$, modulo 2. If M is invertible, then for every nonzero $\beta \in \mathbb{F}_2^{64}$ the function $\langle \beta, S^{(r)} \rangle$ is not balanced over I , nor over any $I' \subseteq I$, for independent round keys.*

Proof. By Proposition 1, applied to the extended function as in the proof of Lemma 5, and by linearity, the coefficient of $x^{1_I} k^{v^{(l)}}$ in $\langle \beta, S^{(r)} \rangle$ is $\sum_j M_{l,j} \beta_j = (M\beta)_l$. Since M is invertible and $\beta \neq 0$, some $(M\beta)_l$ equals 1, and the argument of Lemma 5 shows that the cube sum is not identically zero. The proof of Lemma 6 applies to any output function. $\square$

The second lemma gives matrix entries for output functions that, through the last round, depend on the input of a single S-box. It formalises the choice of last-round key monomials within one S-box that Hebborn et al. [19] use to build their integral-resistance matrices (see also the description by Peng et al. [34]), for an arbitrary component of the S-box and a single last-round key bit. Let $r \geq 2$, let $I \neq \emptyset$, and fix an S-box m of round r . For $i = 0, \dots, 3$ write $s_i = S_{4m+i}^{(r-1)}$ for bit $4m+i$ of the state before round r and $k_{r,i}$ for bit $4m+i$ of RK_r , so that the input of S-box m in round r is $z = s \oplus k_r$ with $z_i = s_i \oplus k_{r,i}$. We identify masks on the four bits of an S-box with subsets of $\{0, 1, 2, 3\}$, and for a nonzero mask u we write $s^u = \prod_{i \in u} s_i$.

Lemma 8 (Last-round component). *Let $f = g(z)$ for a Boolean function g on $\mathbb{F}_2^4$ with ANF coefficients g_w , $w \subseteq \{0, 1, 2, 3\}$. For a key pattern V of the first $r-1$ rounds and a nonzero mask u , let $c_V(u)$ be the coefficient of $x^{1_I} k^V$ in s^u , as a polynomial in the plaintext and the round keys $RK_1, \dots, RK_{r-1}$. Then for every $t \in \{0, 1, 2, 3\}$ the coefficient of $x^{1_I} k^V k_{r,t}$ in f is*

$$\bigoplus_{u \neq 0, t \notin u} g_{u \cup \{t\}} c_V(u).$$

Moreover, $c_V(u)$ is the number, modulo 2, of $(r-1)$ -round monomial trails from $\mathbf{1}_I$ with key pattern V that end in the mask u on S-box m and zero elsewhere.

Proof. Expand $g(s \oplus k_r) = \bigoplus_w g_w \prod_{i \in w} (s_i \oplus k_{r,i})$. The variables s_i do not depend on RK_r , so only the terms with $w \ni t$ contain $k_{r,t}$, and in such a term the only product whose sole factor from RK_r is $k_{r,t}$ is $s^{w \setminus \{t\}} k_{r,t}$. For $w = \{t\}$ this is $k_{r,t}$ alone, which does not contain x^{1_I} because $|I| \geq 1$. Collecting the coefficient of $x^{1_I} k^V$ in $s^{w \setminus \{t\}}$ for all other $w \ni t$ and writing $u = w \setminus \{t\}$ gives the formula. The last statement is Proposition 1 for the extended function $(x, RK_1, \dots, RK_{r-1}) \mapsto s^u$, whose output monomial is the mask u on S-box m ; each key bit enters once, so its trails from $x^{1_I} k^V$ are the trails with key pattern V , as in the proof of Lemma 5. $\square$

5 Validation

Reference implementation. Our Python implementation reproduces all four published test vectors, agrees with the authors’ reference code on 1000 random encryptions, and is cross-checked against a vectorised implementation. Section 10 gives two corrections to the published key-schedule description.

Local models. The S-box table is computed from the ANF. We verified exhaustively the Hu–Yap rule for $n \leq 5$, Lemma 1 for $n \leq 4$ (all 2^{16} exponent pairs), Lemma 2 for $n \leq 8$, and the carry encoding for $n \leq 6$.

Benchmark. With the same SAT machinery and BDP rules we reproduce the results of Eskandari et al. [14] for GIFT-64 [1] and PRESENT [7]:

- GIFT-64: a 9-round property with 63 active bits and no 10-round property. When the constant bit is the most significant bit of a nibble, as in the published result, we find 30 balanced bits; the best position gives 32.
- PRESENT: a 9-round property with 60 active bits and one balanced bit, and no 10-round property.

Soundness against experiment. We evaluated cube sums for random constants and independent random round keys in three settings:

- the word cubes of Table 1, with 1016 trials;
- the 28 cubes of the tightness study in Section 8, with 200 trials per cube and round;
- every cube of dimension at most 16 in Table 3, including those of the variants, with 400 trials.

No certified bit ever had a nonzero sum.

Trail counting. We validated the counting used for presence proofs in two ways. First, for 123 (key pattern, bit) pairs at two rounds and 20 at three rounds (cubes of dimension 5 and 7), the parity of the enumerated count equals the coefficient $a_{\mathbf{1}_I, v}$ computed directly by a Möbius transform of the cipher over the cube and key bits involved; 11 of these coefficients are 1. Second, we repeated the enumeration with blocking clauses over *all* variables of the instance instead of the layer masks. On 31 instances both counts agree: exactly on the 11 instances below the cap, and both exceed the cap on the other 20. This supports the claim that the enumeration counts every trail exactly once.

Checkable artifacts. For the boundary results we produced artifacts that can be verified without trusting the SAT solver. They do not cover the correctness of the CNF generator itself, which rests on the exhaustive checks of the local rules above and on the agreement with experiment. An UNSAT answer is certified by re-solving the CNF with an external CaDiCaL 3.0.1 [5] binary that writes a DRAT proof, which is then checked by `drat-trim` [44]. A SAT answer is certified by converting it into a trail and validating the trail with a checker that re-derives every layer from the cipher specification and the local rules, without the CNF; for a trail that belongs to a presence proof, the checker also confirms its key pattern. A trail count n is certified by validating each of the n trails found, by checking that they are pairwise distinct, and by a DRAT proof that the instance, with the input mask, the output mask and the key pattern fixed by unit clauses, becomes unsatisfiable once one blocking clause per trail found is added.

Table 1: Balanced bits of $S^{(r)}$ for 16-bit word-saturated cubes. For Dipper, the empirically zero bits (1016 trials) coincide with the certified bits in every entry for $r \geq 2$. The last two columns give, for the analysis variants, the last round with empirically zero bits (400 trials) and their number (416 trials: 400 with random round keys, 16 with the real key schedule); these two columns are empirical only. Bold: last round with balanced bits.

Saturated word	$r=1$	2	3	4	5	6	Dipper [⊕]	Dipper [⊘]
A	64	64	64	38	3	0	6 (26 bits)	7 (14 bits)
B	64	56	24	0	0	0	5 (7 bits)	7 (9 bits)
C	64	64	64	38	3	0	6 (25 bits)	7 (4 bits)
D	64	60	25	0	0	0	5 (8 bits)	7 (6 bits)

A blocking clause excludes the layer masks of one trail; with the key pattern fixed, the layer masks determine the trail, so each clause excludes exactly one trail, and the proof shows that the enumeration is complete. We certified in this way:

- certificates: every certified (cube, bit) pair of Dipper at six rounds, of Dipper[⊕] at nine rounds and of Dipper[⊘] at ten rounds, the smallest cubes of Table 3 for Dipper, and the five-round word-cube properties (138 distinct instances; three five-round instances occur both as word cubes and as frontier cubes, so there were 141 proof runs), and every certified bit of the two minimal six-round cubes;
- “no certificate” answers: every one of the 3×4096 (cube, bit) pairs at Dipper seven rounds, Dipper[⊕] ten rounds and Dipper[⊘] eleven rounds (12 288 trails), and the 4456 answers of the minimality search of Section 6;
- presence proofs: the 4142 counts of Sections 6 and 8 (32 620 trails) and the 1791 counts behind the degree lower bounds of Section 7.2 (2187 trails);
- degree upper bounds: all 1856 bounds of Section 7.2;
- every solver answer behind the seven-round linear-combination result (Section 6.4).

All DRAT proofs were verified and all trails passed the check. The remaining UNSAT answers, e.g. the certificates at $r \leq 5$ in Table 2 and the model-comparison instances, rely on the solver answer alone. The CNF hashes, the tool versions and the verification logs are part of the repository.

6 Certified properties of Dipper

6.1 Reproducing the published experiment

Table 1 repeats the published experiment: one 16-bit word is saturated, the other 48 bits are constant, and 1016 trials are run (1000 with independent random round keys, 16 with the real 128-bit key schedule). A bit counts as empirically balanced only if its sum is zero in every trial. As published, five rounds are the maximum, and the round count refers to the state $S^{(r)}$ itself. For every word and every round $r = 2, \dots, 5$, MP-EL certifies exactly the empirically balanced bits, so these properties now hold for all round keys. The number of trials matters. An earlier run of ours with 80 trials reported two more bits for word C , one at $r = 4$ and one at $r = 5$, whose sums turned out to be nonzero with probability of roughly 0.5–1%. Certificates remove exactly this kind of error. In every trial we also confirmed $\bigoplus T^{-1}(S^{(r+1)}) = \bigoplus S^{(r)}$ (Lemma 4).

Table 2: Cubes of dimension 63 (constant bit p ; 64 cubes per row). MP-EL, MP-circuit and BDP give identical results (MP-circuit was run for $r \leq 7$).

Rounds r	cubes with a certificate	max. certified bits	solver time (MP-EL, 64 cubes)
4	64	64	3.9 s
5	64	42	6.1 s
6	14	9	7.9 s
7	0	0	8.6 s
8	0	0	11.1 s
9	0	0	12.2 s

Table 3: Smallest certified cubes found (dimension d , i.e. 2^d chosen plaintexts; greedy search). The last value in each row is minimal within the model. “–”: no certificate for any bit-aligned cube (Lemma 3); blank: not computed.

Rounds r	3	4	5	6	7	8	9	10
Dipper	2	4	16	60	–	–	–	
Dipper [⊕]			7	22	44	59	63	–
Dipper [∅]			3	8	15	48	59	63

6.2 Certified properties over all bit-aligned cubes

By Lemma 3, the 64 cubes of dimension 63 (one constant bit p) decide which rounds admit a certificate for some bit-aligned cube. Table 2 summarises them. At six rounds, certificates exist for 14 of the 64 positions of the constant bit, and the best positions $p \in \{4, 6, 7\}$ give nine balanced bits, $\{1, 18, 19, 24, 41, 48, 51, 56, 58\}$. At seven, eight and nine rounds no position gives a certificate. Within the model, therefore, no bit-aligned cube of any dimension certifies a single output bit at seven rounds or more. Every one of the 4096 seven-round queries returned a trail that passed the independent check. With Lemma 4, the six-round properties give seven-round distinguishers on $T^{-1}(C)$.

Data complexity. Starting from certified cubes, we removed active bits one at a time as long as some bit stayed certified, restarting from several cubes and random orders. Table 3 lists the smallest certified cubes *found* in this way. They are upper bounds on the data required. At five rounds the smallest cube found has size 2^{16} , the size of the published word-saturation structure (word C). At six rounds the minimum can be proven. If a cube is certified for bit j , then by Lemma 3 so is every 63-dimensional cube containing it. Its constant bits therefore lie among the positions p whose 63-dimensional cube certifies bit j , a subset of the 14 positions with a six-round certificate (Table 2). For each bit j , the sets of constant bits of cubes certified for j are closed under taking subsets, again by Lemma 3, so they can be searched level by level. The largest such set, over all j , has four elements. Hence 2^{60} is the minimum within the model, and exactly two cubes attain it: the cube with the nibble $\{4, 5, 6, 7\}$ constant, which certifies six bits, and the cube with the nibble $\{12, 13, 14, 15\}$ constant, which certifies bits $\{24, 51, 56, 58\}$. Each of the 4456 “no certificate” answers of this search was turned into a trail and validated independently. The same search shows that no cube of dimension below 63 is certified for Dipper[⊕] at nine rounds or for Dipper[∅] at ten rounds.

6.3 Seven rounds: presence proofs

The seven-round entries of Table 2 say only that the model has a trail. For each of the 64 cubes of dimension 63 and each of the 64 output bits of $S^{(7)}$ we searched for a presence proof (Lemma 5),

and found one in all 4096 cases. The trail counts of the key patterns found range from 1 to 1397, and 1958 of them equal 1. For every proof all trails were enumerated again and passed the independent check, and a DRAT proof certifies that the enumeration is complete (Section 5). The search took about 8 core-hours in total. Reusing patterns across positions of the constant bit (Section 4.3) was decisive: whole patterns gave 605 of the 4096 proofs and partial patterns 2107, and the random search gave the remaining 1384. Every bit-aligned cube of dimension at most 63 lies in one of the 64 cubes, so by Lemma 6 no bit-aligned cube has a balanced output bit after seven rounds, for independent round keys. Since the sum of $T^{-1}(S^{(8)})$ over a cube equals that of $S^{(7)}$ (proof of Lemma 4), the same holds for $T^{-1}(S^{(8)})$. The seven-round answer of the model is therefore exact for independent round keys. Over bit-aligned cubes and single output bits, the six-round properties of Table 2, and their extension to $T^{-1}(C)$ by Lemma 4, are the longest key-independent integral properties of Dipper with independent round keys. This does not exclude probabilistic or weak-key distinguishers.

6.4 Seven rounds: linear combinations of output bits

An attacker may also sum a linear combination $\langle \beta, S^{(7)} \rangle$ of output bits. With Lemma 7 we show that no nonzero combination is balanced after seven rounds over any bit-aligned cube. For each of the 64 cubes of dimension 63 we need 64 key patterns $v^{(0)}, \dots, v^{(63)}$, one per row of M , for which M is invertible.

Deciding invertibility. The matrix M need not be computed in full. Let G be the graph on the output bits with an edge $l \rightarrow j$ ($j \neq l$) whenever pattern $v^{(l)}$ has at least one trail to bit j ; without an edge, $M_{l,j} = 0$. We order the strongly connected components of G so that every edge between different components goes to an earlier component, and permute the rows and the columns of M by the same order. Then M is block lower triangular, with one diagonal block per component, and it is invertible if and only if every diagonal block is. A component $\{l\}$ with a single bit needs $M_{l,l} = 1$, i.e. $v^{(l)}$ must be a presence proof for bit l . Inside a larger component a row need not prove its own bit; only the invertibility of the block matters. A block is decided by counting the trails of all its entries or, where this is infeasible, with Lemma 8. The lemma applies to output bits of $S^{(7)}$ that come from a retained word of the last ARX layer: through the rotation and the bit permutation, such a bit is an output bit of one S-box m of round 7, i.e. a function of the S-box input z alone, and so is the sum of two such bits of the same S-box. It gives the entry of a row whose pattern consists of a six-round pattern V and a single key bit $k_{7,t}$ of S-box m . In the final matrices, every block that could not be counted consists of two such bits a, b of one S-box, and one of its rows is $(1, 1)$ by counting. Its determinant is then the sum of the two entries of the other row, i.e. the entry for $S_a \oplus S_b$, which Lemma 8 gives from the ANF of this component of the GIFT S-box and six-round counts $c_V(u)$. We call this the *sum argument*.

Finding the patterns. We started from the 64 presence proofs of Section 6.3 and replaced the patterns of bits on cycles of G by other presence proofs with no trail to the rest of their cycle. For the cube with constant bit 0, almost every pattern had trails to a common set of twelve output bits, all high bits of the two added words, and the cycles that remained were small. Their blocks were often singular, or their entries could not be counted. For two output bits 0 and 1 of one last-round S-box of a retained word there is a structural reason. The two bits have the same ANF except for the constant and the monomials z_1 and $z_0 z_2$; their sum is the component $1 \oplus z_1 \oplus z_0 z_2$ of degree two, and a trail from a nonzero mask never enters an S-box in the empty monomial. A trail to one of the two bits whose last-round S-box input is any other monomial is also a trail to the other bit, with the same key pattern, so the two entries of a row can differ only through trails that enter the last S-box in z_1 or $z_0 z_2$. A similar dependence of trail parities on the last-round S-box input monomials underlies the interfering monomials

that Peng et al. [34] found in PRESENT. Pairs formed by a bit $(X \boxplus Y)_i$ of an added word and the retained bit Y_i behaved similarly, which may be due to their common linear term Y_i ; we did not analyse this further. We separated such pairs by patterns that use the last round: a six-round presence proof V for one input bit of the S-box, extended by a single last-round key bit t . For S-box output bits 0 and 1 and a presence proof V for s_0 , for instance, Lemma 8 gives the row $(1, 1)$ for $t = 1$ and the row $(0, 1)$ for $t = 2$, whatever the other counts $c_V(u)$ are. Indeed, the monomials that contain z_1 and another variable are z_0z_1 in both bits, and those that contain z_2 and another variable are none in bit 0 and z_0z_2 in bit 1, so in both cases only $u = \{0\}$ contributes. These two rows have determinant 1. How the patterns were found does not affect the result, which we verified independently.

Independent verification. A separate program starts from the final patterns only. For each cube it recomputes G and its components, checks that every edge goes to an earlier component, and decides every diagonal block as described above; it validates every trail it counts with the independent checker, including the key pattern. Over the 64 cubes the matrices have 4016 components: 3948 single bits, each with a counted diagonal entry 1, and 68 blocks, of which 57 have two bits, 10 three and one four. All 68 blocks are invertible: 47 by counting all their entries, and 21 blocks of two bits, in 13 of the cubes, by the sum argument. In all 21 the two bits are S-box output bits 0 and 1 and the pattern of the other row ends in the key bit $t = 0$. The only monomial of $1 \oplus z_1 \oplus z_0z_2$ that contains z_0 and another variable is z_0z_2 , so the determinant is the single six-round count $c_V(\{2\})$. The counted entries rest on 51 698 seven-round trails and the sum arguments on 55 six-round trails. The block-triangular structure uses 128 931 zero entries outside the diagonal blocks, and a few zero entries lie inside the blocks. Every one of these solver answers is certified by a checked DRAT proof (Section 5): for each row, one proof that its pattern has no trail to any output bit without an edge (4096 proofs, covering all 194 321 zero entries of the 64 matrices), a proof that the enumeration is complete for each of the 4174 counted entries, and the same for each of the 21 six-round counts of the sum arguments. All 8291 proofs were verified.

By Lemma 7, no nonzero linear combination of the output bits of $S^{(7)}$ is balanced over any bit-aligned cube, for independent round keys, and by Lemma 4 the same holds for $T^{-1}(S^{(8)})$. This is the integral-resistance property of [19] after seven rounds, restricted to bit-aligned cubes. Since M is invertible, it also implies once more that every single output bit is unbalanced.

6.5 Where the balanced bits come from

We map the nine balanced bits at the six-round boundary back through the final bit permutation. Every one of them is an output of a retained word in the last round: bits $D_3, D_4, D_5, D_6, D_{11}$ and $B_4, B_{12}, B_{13}, B_{14}$. The same holds for the five-round word-cube properties. Bits $\{51, 56, 58\}$ of the word- C cube map to D_3, B_4, B_{14} , and bits $\{1, 19, 24\}$ of the word- A cube map to D_5, D_{11}, B_{12} . No output bit of the last-round additions is certified at the boundary. This is an observation about the certified bits, not a claim about every output bit of the additions: bit 0 of $A \boxplus B$, for example, is linear. Section 7.2 relates it to degree.

7 The role of the modular addition

7.1 Ablation

Table 4 compares Dipper with $\text{Dipper}^\oplus$ and Dipper° . The three ciphers were analysed under identical conditions: all 64 maximal cubes, MP-EL and BDP (which again agree), and the same solver settings (conflict budget 10^5 with a $5 \cdot 10^6$ retry; no instance was left unresolved). Within the model, the two additions shorten the longest certified property from nine rounds (XOR) or ten rounds (no addition) to six. The empirical word-cube experiment (Table 1) orders the three

Table 4: Longest round count with a certificate for some bit-aligned cube (all cubes, by Lemma 3) and the maximum number of balanced bits at that round.

Variant	longest certified r	max. certified bits	with free round (Lemma 4)
Dipper ($\boxplus$)	6	9	7
Dipper $^\oplus$	9	5	10
Dipper $^\circ$	10	2	11

ciphers the same way: 5, 6 and 7 rounds. The data needed shifts in the same direction. For six rounds the smallest certified cubes found have size 2^{60} for Dipper, 2^{22} for Dipper $^\oplus$ and 2^8 for Dipper $^\circ$. Figure 2 shows this round by round.

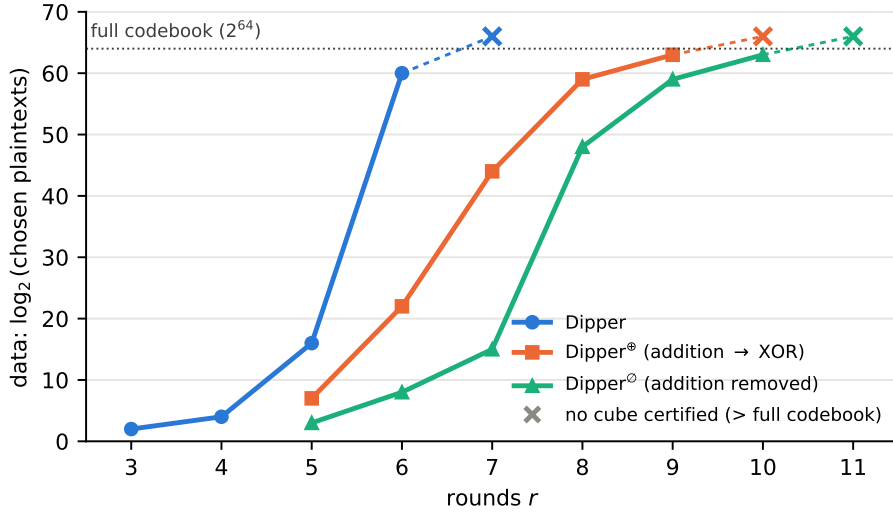


Figure 2: Data needed by the smallest certified cube found, per round, for Dipper and the two variants without carries (values of Table 3). A cross, joined to the last point by a dotted line, marks the first round at which no bit-aligned cube has a certificate; it is drawn above the full codebook because no amount of data suffices there. Dipper’s curve reaches the full codebook three to four rounds earlier.

The variants keep the rotations, the S-box, the permutation and the full-state key addition, so they are not GIFT-64. A difference measured with an existence-based model describes certified properties. It describes the true algebraic behaviour only where the model is tight. We measured tightness only on Dipper for up to five rounds (Section 8).

7.2 Algebraic degree

By Section 3, a bit of degree less than d is balanced over every cube of dimension d , so degrees give a global view of integral resistance. We consider the degree in the plaintext bits, with the round keys as independent variables. For every output bit j and round count r we first computed the smallest D for which MP-EL with a free input mask of weight at least $D + 1$ and output e_j is unsatisfiable. This certifies the upper bound $\deg_x S_j^{(r)} \leq D$ for all round keys. We then looked for a matching lower bound: a monomial x^u with $\text{wt}(u) = D$ taken from the model, and a presence proof for it (Lemma 5). At the degree limit the first-round part of the key monomial is often forced to be empty, so the search also uses patterns with $v_1 = 0$. For one one-round bit the search failed and we computed its full ANF instead, which has 32 variables; the ANF of four further one-round bits agrees with their presence proofs. All 64×7 cases for Dipper and

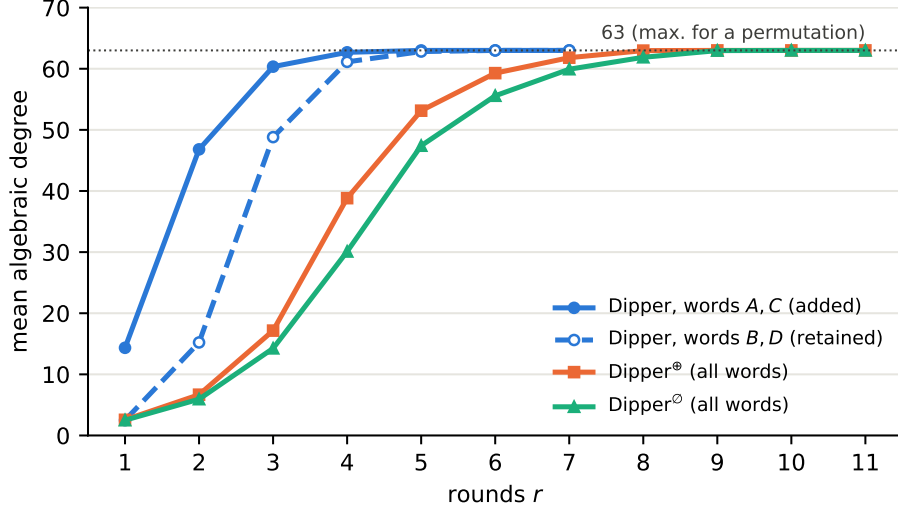


Figure 3: Mean algebraic degree of the output bits of $S^{(r)}$, with the round keys as independent variables. The plotted values are proven lower bounds. They equal the certified upper bounds in 1843 of the 1856 cases and differ by at most two otherwise. For Dipper, bits are grouped by the word of the last ARX layer that produced them.

Table 5: Algebraic degree after one round (exact), by bit position i in the output of the ARX layer.

i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$A \boxplus B$	3	5	7	8	9	11	13	15	16	18	20	21	22	24	25	26
$C \boxplus D$	2	4	6	6	8	10	12	12	14	16	18	18	20	22	24	24
B (retained)	2	2	3	3	2	2	3	3	2	2	3	3	2	2	3	3

64×11 for each variant were treated without timeouts.

The two bounds coincide in 1843 of the 1856 cases, so there the degree is exact. The other 13 cases lie in Dipper at rounds 2 and 3 and in Dipper[⊕] at rounds 4 to 6. In 12 of them the lower bound is one below the upper bound, and in one it is two below. Here the upper bound may be loose because trails cancel. Every upper bound is certified by a checked DRAT proof, and every lower bound by a presence proof with a certified count, except for the one-round bit whose degree we took from its ANF (Section 5).

Figure 3 and Table 5 support three observations.

The addition acts immediately. After one round, the degree of bit i of $A \boxplus B$ grows almost linearly in i , from 3 to 26, the pattern predicted by Lemma 2 and Corollary 1. The retained word B stays at the S-box degree 2 or 3. The growth is slower than $3(i+1)$ because neighbouring bits share S-boxes.

Retained words lag by about one round. The mean degree of the added words reaches 60 at $r = 3$, and the retained words need one more round. After five rounds every added-word bit has degree 63, while six retained-word bits, $\{1, 19, 24, 51, 56, 58\}$, have degree exactly 62. Degree 62 certifies the bit over every 63-dimensional cube. These six bits are exactly the bits certified for all 64 maximal cubes at five rounds, and the same six bits carry the six-round certificate of the 2^{60} cube (Table 3).

The degree saturates three to four rounds earlier with the additions. From $r = 6$ on, every output bit of Dipper has degree 63; for Dipper[⊕] and Dipper[⊙] this happens only from $r = 9$ and $r = 10$ on (the last rounds with a degree below 63 are 5, 8 and 9, respectively). The ordering

matches Table 4, but the degree does not determine the certificate boundary: at six rounds Dipper still has certificates for particular 63-dimensional cubes although every degree is 63, and Dipper[⊕] at eight rounds has 64 certified bits on some cube while only two bits have degree below 63. Certificates are per cube, the degree is over all cubes (Remark 1). In the linear variants the degree can grow only through the S-boxes, at most by a factor of three per round. In Dipper each addition contributes the additional factor of Corollary 1. Because the degrees are exact, these are statements about the ciphers with independent round keys, not only about the model.

8 Precision of the models

MP-EL versus BDP. On all 64 maximal cubes, MP-EL and BDP produced identical certificate sets: for $r = 4, \dots, 9$ on Dipper, up to $r = 10$ on Dipper[⊕] and up to $r = 11$ on Dipper[⊘]. MP-circuit agreed with both wherever it was run ($r \leq 7$). On 140 further (cube, round) instances of dimension 4–16 on Dipper, all three models produced identical certificate sets, at similar cost (8.4s, 8.8s and 6.2s in total). This shows that the two *existence* models agree on the tested instances. It does not show that cancellation-aware monomial prediction would have no advantage over BDP in general; for the instances of Table 6, the end of this section shows that it has none. Part of the agreement is structural. Every S-box layer of Dipper is preceded by a full-state key addition. In MP-EL the composition “key addition, then S-box” allows exactly the transitions $u \rightarrow v$ with $S[w \rightarrow v] = 1$ for some $w \succeq u$, which is precisely the BDP S-box table. The two models can therefore differ only inside the ARX layer, and there we observed no difference.

Certificates versus experiment. Table 6 compares certificates with experiment on 28 cubes of dimension 4–16: word-aligned, nibble-aligned and random. A bit is *empirically zero* if its sum is zero in 200 trials, and it is a *gap* bit if it is empirically zero but not certified. Gap bits were re-tested with 3000 more trials (1000 for 16-dimensional cubes); *persistent* gaps stayed zero. The last column counts the persistent gap bits that we proved unbalanced.

- At four and five rounds the certificates match experiment up to three gap bits, none of them persistent.
- At two and three rounds, about 7–17% of the empirically zero bits are not certified, and 46 bits stay zero in every re-test.

Such a bit is either an exact property lost to cancellation (Example 1) or a sum that is nonzero only with small probability. We tried two exact methods on these bits.

- *Trail parity per key monomial.* For bit 30 of a 12-dimensional random cube at two rounds, the number of trails exceeded our enumeration cap of 10^5 . The key monomial was that of a single trail and contained few key bits.
- *Low-weight evaluation.* The cube sum g is a multilinear polynomial in the key bits. If it has degree at most D , it vanishes identically iff it vanishes on every point of Hamming weight at most D . MP-EL gives the key support of g and a bound on D . For the 46 persistent gap bits the support has 33–88 key bits and the degree bound is 23–65. For 43 of them the number of low-weight points exceeds 10^{15} . The three smallest cases (support 33–38, degree bound 23–27) need about $8.5 \cdot 10^9$ – $2.7 \cdot 10^{11}$ points. Within a time limit of 800s per case we evaluated 2.1–3.8 million points, covering all points of weight at most 5 or 6. No nonzero cube sum appeared, but the enumeration is far from complete, so this method decided none of these bits.

Presence proofs with maximal key patterns (Section 4.3) decide all of them. For each of the 46 persistent gap bits we found a key pattern with an odd count, between 1 and 9 trails, and

Table 6: Certified versus empirically zero bits over 28 cubes of dimension 4–16 (sums over all cubes). Last column: persistent gap bits proved unbalanced by a presence proof (Lemma 5).

Rounds	certified	empirically zero	gap	persistent gap	proved unbalanced
2	1283	1384	101	21	21
3	446	537	91	25	25
4	91	93	2	0	–
5	6	7	1	0	–

validated the trail; the search took 11 s in total. None of these bits is balanced. Each of their cube sums is a nonzero polynomial in the round keys that vanished in all 3200 trials. This is the failure mode of the 80-trial experiment of Section 6 in a stronger form: uniformly random keys almost never reveal it, and an algebraic argument is needed to decide it. The presence proof also shows where to look. If v is the key pattern of a proof, some round key whose bits lie inside v gives the cube sum 1, because the sum of the cube sum over all such keys equals the coefficient of k^v . Sampling sparse keys inside v found such round keys for 31 of the 46 bits, and each was confirmed with the reference implementation. For these bits the claim can be checked without the SAT model. Every other bit that is not certified had a nonzero sum in some trial. Hence every bit of the 112 instances in Table 6 is decided, and on every instance the certified bits are exactly the balanced bits. On these instances the existence model loses nothing to cancellation, and a model that counts trails could not certify more. The frontier itself (six and seven rounds, cubes of dimension 60–63) is out of experimental reach; Section 6.3 treats it with presence proofs.

9 A partial-key filtering extension

This section describes how the six-round properties could be used for key filtering over one further round. We present it as a proposed procedure with an explicit cost model. It is not a validated eight-round attack.

Procedure. Let bit j of $S^{(6)}$ be certified over a cube I . For eight-round ciphertexts C , Lemma 4 gives

$$\bigoplus_{x \in \mathcal{C}_I(c)} T^{-1}(T^{-1}(C) \oplus RK_8)_j = 0.$$

Bit j of $T^{-1}(z)$ depends on four bits of z if its nibble lies in word B or D , where the inverse layer is only a rotation. It depends on up to 32 bits if the nibble lies in A or C , because of the borrow chain of the subtraction. For $j = 1$, which is certified over the 2^{60} cube, it depends on four bits of $Z = T^{-1}(C)$ and hence on four bits of RK_8 . One structure of 2^{60} chosen plaintexts is processed in a streaming fashion:

1. For each ciphertext, compute $Z = T^{-1}(C)$ (keyless) and toggle one of 16 parity bits indexed by the four relevant bits of Z .
2. Let $h : \mathbb{F}_2^4 \rightarrow \mathbb{F}_2$ be the partial inverse that maps the four relevant bits of z to bit j of $T^{-1}(z)$ (inverse rotation of word D , then the inverse S-box of the corresponding nibble). For each of the 16 guesses g , evaluate $\bigoplus_{v: \text{parity}[v]=1} h(v \oplus g)$ over the 16 table entries, and discard g if the result is 1.

One structure costs 2^{60} evaluations of T^{-1} plus 2^8 table operations. It needs 16 bits of memory besides the ciphertext stream. Every structure uses a fresh value of the four constant bits.

Assumptions and small-scale evidence. The right guess always passes. The number of wrong guesses left after N structures depends on how the wrong-guess sums behave. Under the usual heuristic that they are independent and uniform, N structures leave $15 \cdot 2^{-N}$ wrong guesses on average. We ran the same procedure at small scale, on six rounds: the 4-round certificate of the cube $\{60, \dots, 63\}$ on bit 1, followed by the free round and one key-guessed round, with 200 random keys. The right guess always survived. A wrong guess survived one structure with probability about 0.565, so the uniform heuristic is optimistic, and eight structures left 1.13 candidates on average, with a unique survivor in 88.5% of the runs. We have not established that this rate carries over to eight rounds or that the structures act as independent filters there. Extrapolating anyway, about eight structures, i.e. 2^{63} chosen plaintexts, would be needed to identify four bits of RK_8 . This recovers round-key bits, not the master key, and the data requirement is half the codebook. We therefore regard the procedure as evidence of a large margin, not as an attack on Dipper.

10 Specification of the analysed cipher and corrections

The results of this paper depend only on the round function, because every certificate holds for arbitrary independent round keys. The round function is fixed by Section 3 and confirmed by all published test vectors, which exercise all 28 rounds.

We implemented both key schedules independently and compared them with the published test vectors and with the authors’ reference implementation (Specification v1.1 and its Python reference code, which generated the vectors of [26]). Our implementation agrees with the reference code on all four published vectors and on 1000 random encryptions (500 per key size). The specification and the reference code are consistent with each other. Two statements in the key-schedule section of the published paper [26] are not, and we correct them here.

1. *Round-key window of Dipper-64/128.* Equation (7) of [26] states $RK_r = K^{(r)}[127:64]$. The specification and the reference code use $RK_r = K^{(r)}[63:0] = k_3 \| k_2 \| k_1 \| k_0$ for both key sizes, and this produces the published vectors.
2. *S-box positions of Dipper-64/96.* Equation (9) of [26] applies the S-box to the top nibble [15:12] of words k'_1, k'_2, k'_4, k'_5 . The specification and the reference code apply it to the key-state bits $K[95:92]$, $K[71:68]$, $K[47:44]$ and $K[23:20]$, i.e. to $k'_5[15:12]$, $k'_4[7:4]$, $k'_2[15:12]$ and $k'_1[7:4]$. For k'_4 and k'_1 the S-box therefore acts on the second-lowest nibble, not on the top nibble. The comments of the specification call these positions “top nibbles”, which is how the error entered the paper. With this correction the published Dipper-64/96 vectors are reproduced. The word permutation, the rotations and the round-constant position $K[52:48]$ are as stated in [26].

The Dipper-64/128 S-box positions (k'_7, k'_3, k'_1 , top nibbles) are stated correctly in [26]. Appendix A lists all published vectors with intermediate round keys and states. Neither correction affects the integral results, which hold for arbitrary independent round keys, or the round function, which the published text states correctly.

11 Discussion

Security implications. The integral properties certified here reach six of Dipper’s 28 rounds on the state, and seven with the keyless final-round inversion. The filtering extension of Section 9 would need half the codebook to reach eight rounds. Integral attacks therefore leave a large margin. This is consistent with the differential, linear and impossible-differential bounds of [26]. None of these results is a proof of security.

Design implication. The ablation and the degree bounds point the same way. With the two half-state additions, the certifiable integral properties end three to four rounds earlier than for an otherwise identical round without carries. Within the state, the retained operands are the part whose degree bound grows most slowly, and every certified property at the boundary lives there. A designer who wants integral saturation to happen sooner could therefore target the retained words, for example by alternating which words are added in consecutive rounds. We have not evaluated this change.

Towards exact bounds. Presence proofs close the gap between “no certificate” and “not balanced” wherever they succeed: for all 46 persistent gap bits and for every output bit of all 64 seven-round cubes of dimension 63. The method goes back to Hebborn et al. [18, 19]. Zeng and Tian [47] adapted it to ciphers without a whitening key, and Peng et al. [34] showed that interfering monomials can make every count even when the key monomial is badly chosen. For Dipper the choice of the key pattern decided the outcome. Taking the key monomial of a single trail gave no odd count within $2 \cdot 10^4$ trails on two seven-round instances, while maximal key patterns gave an odd count for every bit we tried. With them the seven-round statement becomes a statement about the cipher for all bit-aligned cubes; the search took about 8 core-hours. A stronger goal is integral resistance in the sense of [19], a guarantee that covers input sets other than bit-aligned cubes and all nonzero linear combinations of output bits. Section 6.4 establishes it for bit-aligned cubes after seven rounds; other input sets remain open. Since RK_1 acts as a whitening key, the theorem of Hebborn et al. would cover them, but it needs matrices over all 64 input monomials of degree 63 jointly. With our presence patterns, the matrix of every single output bit over these 64 monomials is singular: two input monomials that differ only inside one nibble usually give equal rows. Hebborn et al. avoid this with key monomials chosen right after the first S-box layer; in Dipper that key is added only after the first ARX layer. We leave this to future work. The framework has been extended to key whitening by modular addition [3] and to complex linear layers [15]. To our knowledge it has not been applied to a round function that itself contains modular additions, as Dipper’s does.

The real key schedule. All presence results assume independent round keys. To test whether they carry over to the key schedules, we extended MP-EL by the key schedule: the master-key bits are variables, each round key is a COPY of the 64 low bits $K^{(r)}[63:0]$ of the key state, and the key update is modelled by wiring, the key-schedule S-boxes and the round constants. A trail then carries a master-key monomial. For the cube with constant bit 0 and six output bits we took 20 samples per bit, key size and round count, alternately maximal and minimal master-key monomials found greedily (as in Section 4.3; a minimal monomial is built by excluding key bits as long as a trail survives), and counted their trails up to a cap of 2000. At six rounds, 44 of the 45 counts below the cap were even for the 128-bit key, and one was odd (15 trails; we did not validate these trails independently, because our checker does not model the key schedule); 75 samples exceeded the cap. For the 96-bit key, 39 counts were even, none odd, and 81 samples exceeded the cap. At seven rounds, 226 of the 240 samples for the two key sizes exceeded the cap and the other 14 counts were even. A possible reason is that each master-key bit enters several round keys, so that trails may occur in pairs that differ only in the round key through which a master-key bit enters; we have not verified this. Proving presence with the key schedule needs a different technique, such as the nested monomial prediction developed for superpoly recovery in cube attacks [21]. We leave it open.

Limitations.

- The seven-round statement concerns bit-aligned cubes, for which it covers every nonzero linear combination of output bits. It says nothing about input sets that are not bit-aligned cubes.

- Independent round keys make the certificates stronger, since they hold for all keys. For the same reason they may miss properties that depend on the key schedule.
- Below the boundary rounds, the data complexities are those of the smallest certified cubes found by a greedy search; at the boundary rounds they are proven minimal within the model.
- A presence proof excludes a key-independent zero sum. It does not exclude a probabilistic or weak-key distinguisher: a cube sum that is a nonzero polynomial can still vanish for most keys, as the gap bits of Section 8 show.
- Presence proofs hold for independent round keys. A cube sum that is a nonzero polynomial in independent round keys could, in principle, vanish for all round keys produced by the key schedule; we do not exclude this.
- The artifacts of Section 5 certify the solver’s answers for a given CNF. That every monomial trail is a solution of the CNF, and hence that no trail is missing from a count or from an unsatisfiable instance, rests on the local rules, which we verified exhaustively for small word sizes, and is supported by the agreement of the counts with the ANF at two and three rounds.
- Degrees are exact in 1843 of 1856 cases; in the other 13 we give an interval of width at most two.
- The filtering extension relies on an extrapolated survival rate.

12 Conclusion

The five-round integral properties that were found experimentally for Dipper now hold for all round keys, backed by checked DRAT proofs. Certified properties extend to six rounds, with 2^{60} data, the minimum for a certified cube within the model, and to seven rounds with the keyless final-round inversion. No bit-aligned cube gives a balanced bit of $S^{(7)}$: the model has no certificate there, and presence proofs show that every output bit, and every nonzero linear combination of output bits, is unbalanced over every such cube, for independent round keys. The two half-state modular additions account for the difference between six certified rounds and the nine or ten rounds of otherwise identical rounds without carries. Only retained-operand bits are certified at the boundary. This is consistent with the exact degree of the addition output bits and with the degree of every output bit and round, which we determined exactly in 1843 of 1856 cases and which grows fastest on the added words. On every Dipper instance we examined, the existence model with exact local transitions certifies the same bits as the conventional division property, partly because a full-state key addition precedes each S-box layer. Presence proofs also decide every persistent gap bit of the 112 tightness instances: none of them is balanced. What remains of integral resistance in the sense of [19] after seven rounds are input sets other than bit-aligned cubes.

Data and code availability

The reference implementation, SAT models, scripts, raw JSON results, the proof-checking pipeline and the sources of this paper are available at <https://github.com/AlexGuseinov/dipper-integral>. The CNF files and DRAT proofs of the boundary certificates (about 60 MB) are provided as a release archive of the repository. The other DRAT proofs of Section 5 are regenerated and checked on the fly; the repository contains their verification results. The DRAT proofs certify the solver’s answers for the generated CNF files; the CNF generator itself remains

part of the trusted base (Section 5). `make test` runs all validation checks, `make reproduce` regenerates every table and figure, and `make certificates` regenerates and checks all proofs and trails.

A Test vectors and intermediate values

All values in hexadecimal, most significant bit first. Dipper-64/128 with $RK_r = K^{(r)}[63:0]$ (Section 10).

	Vector 1	Vector 2
K	000102030405060708090A0B0C0D0E0F	FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
P	0123456789ABCDEF	FFFFFFFFFFFFFFFF
RK_1	08090A0B0C0D0E0F	FFFFFFFFFFFFFFFF
$S^{(1)}$	87D2D72CD66C68F2	3118311831183108
RK_2	140506076303E0F0	EEEEEEEEEEEEEEEE
$S^{(2)}$	41C18243B4605E18	01A6DC1CB8EEB9D1
RK_3	1001020208C00F0E	EEEEEEFEEBFFFF
$S^{(3)}$	22C358EDC3BD1695	F49F4481BC92B880
RK_{28}	78090A148A9BF0E0	FFFFFEE02664FFFF
$C = S^{(28)}$	4B46284387969060	6ABB4518063706B0

Dipper-64/96 under the corrected S-box positions (Section 10).

	Vector 1	Vector 2
K	00112233445566778899AABB	000000000000000000000000
P	0123456789ABCDEF	0000000000000000
RK_1	445566778899AABB	0000000000000000
$S^{(1)}$	97B834DFA857A809	3118311831183108
RK_2	6227BBBA00A12233	0001100000100000
$S^{(2)}$	E77DDB5526AFBAB0	B23376D0F8957AFC
RK_3	402A433264F56697	0006100010100010
$S^{(3)}$	3BDC9FD0104AC1C2	11D992CE930C3154
RK_{28}	200AEA1FD9DD001D	E5659977E0A29797
$C = S^{(28)}$	5F189202D3152B8F	D873355EC63B1B4B

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